SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology
Department of Agricultural Engineering

AGE 219: Basics of Computer Programming

Official Course Syllabus and Learning Contract

Course Instructor:

Dr. Kadeghe Fue, PhD, P.Eng (T)

Target Audience: Engineering Students (Agricultural, Irrigation, Post-Harvest, and related fields)
Primary Tools: Python 3.x and Google Colaboratory
Academic Year: 2025/2026

This syllabus outlines the roadmap for transitioning from traditional engineering to Digital Agriculture (AgriTech). It serves as a formal agreement on the course objectives, content, and ethical standards.

1 Course Overview

This course introduces fundamental computer programming concepts to engineering students, emphasizing practical problem-solving in the agricultural and biosystems engineering sectors. By utilizing Python and Google Colab, students will learn how to write algorithms, handle data, and build simulations without the need for complex local software installations. The course bridges the gap between traditional engineering mechanics and modern digital agriculture (AgriTech).

2 Recommended References

2.1 Books

1. **"Python Crash Course: A Hands-On, Project-Based Introduction to Programming"** by Eric Matthes. *(Foundational programming logic)*.
2. **"Python Programming and Numerical Methods: A Guide for Engineers and Scientists"** by Qingkai Kong, Timmy Siauw, and Alexandre Bayen. *(Computational mathematics for engineers)*.
3. **"Automate the Boring Stuff with Python"** by Al Sweigart. *(Practical scripting for agricultural research datasets)*.

2.2 Websites

1. **Official Python Documentation (`docs.python.org`):** The definitive standard reference.
2. **W3Schools Python Tutorial (`w3schools.com/python`):** Interactive syntax lookups.
3. **Stack Overflow (`stackoverflow.com`):** The community forum for troubleshooting engineering algorithms.

3 Detailed Lecture Schedule

Lecture 1: Introduction to Programming and Google Colab

Topics: Definition of programming; The role of code in Agriculture 4.0; Python environment setup; Understanding Markdown vs. Code cells.
Case Study: Writing the first "Hello, Farm!" program and executing basic math.

Lecture 2: Variables, Data Types, and Basic Operators

Topics: Naming conventions (PEP 8); Integers, Floats, Strings, and Booleans; Arithmetic and Comparison operators.
Case Study: Calculating fertilizer application rates (kg/ha) for maize production.

Lecture 3: Control Flow (Decision Making)

Topics: `if`, `elif`, and `else` statements; Logical operators (`and`, `or`, `not`).
Case Study: A smart irrigation simulator based on real-time soil moisture thresholds.

Lecture 4: Iteration and Loops

Topics: `for` loops and `while` loops; `break` and `continue` statements; Iterating through datasets.
Case Study: Processing 7-day cumulative rainfall data for crop water requirement analysis.

Lecture 5: Functions and Modular Programming

Topics: Defining functions (`def`); Passing arguments and return values; Python standard math module.
Case Study: A custom function to calculate the draft force of a plow based on soil type and width.

Lecture 6: Data Structures (Lists and Dictionaries)

Topics: Indexing and slicing lists; Key-value pairs in Dictionaries; Mutability.
Case Study: Organizing real-time IoT greenhouse data (temperature, humidity, CO_2).

Lecture 7: File Handling and Data Import

Topics: Reading and writing `.csv` files; Uploading local files to the Colab cloud environment.
Case Study: Identifying frost events in an orchard from a historical weather station CSV.

Lecture 8: Introduction to Object Oriented Programming (OOP)

Topics: Classes and Objects; Attributes and Methods; The `__init__` method; Principles of encapsulation in engineering models.
Case Study: Modeling agricultural machinery as digital objects (e.g., a `Tractor` class with methods for `move` and `consume_fuel`).

Lecture 9: Data Analysis and Visualization (Pandas & Matplotlib)

Topics: Introduction to DataFrames; Data cleaning and filtering; Line, bar, and scatter plots; Professional formatting of engineering graphs.
Case Study: Analyzing machinery telematics data to visualize the relationship between tractor engine load and fuel efficiency.

Lecture 10: Final Mini-Project Integration

Topics: Full-stack integration of logic, functions, data structures, and visualization.
Case Study: "The Smart Grain Bin Manager" — An automated system to prevent spoilage.

4 Academic Honesty Clause

In the School of Engineering and Technology, professional integrity is paramount. Engineering solutions affect public safety and farm livelihoods; therefore, the work you submit must be your own.

- **Originality:** Students are expected to write their own logic. Copying code from classmates or external sources without attribution is considered academic misconduct.
- **Logic Comprehension:** If you use a snippet of code found online, you must be able to explain every line of that code during an oral evaluation.
- **Collaboration:** While group discussion is encouraged, the final implementation of your assignments and project code must be performed individually.

5 Responsible Use of Artificial Intelligence (AI)

As we embrace Agriculture 4.0, tools like ChatGPT and Gemini are part of the modern engineer's toolkit. However, they must be used responsibly:

- **AI as a Tutor, Not a Replacement:** You may use AI to explain complex concepts or debug errors, but using AI to generate entire assignments is prohibited.
- **Verification Mandate:** AI can produce "hallucinations" or mathematically incorrect code. It is your responsibility as an engineer to verify that AI-suggested algorithms follow physical laws and safety standards.
- **Disclosure:** If AI was used to assist in debugging or structuring a project, a "Declaration of AI Use" must be included in the comments of your Python notebook.
- **The Safety Factor:** Never trust AI-generated code for safety-critical systems (e.g., autonomous machinery or chemical dispensers) without rigorous manual testing.

Document Certified by the Department of Agricultural Engineering